REFERENCE GUIDE • STATISTICS.PDF (270 PAGES)

Sequential Table of Contents & Topic Index

Complete chronological mapping of modules, section headings, subtopics, and page ranges

Document Overview: This index preserves the exact sequential order of all headings and instructional topics appearing across the 270 pages of *Statistics.pdf*, organized under the four primary curriculum modules defined in the source slides.

#	Heading / Topic	Key Subtopics & Concepts Covered	Page Range
Module 1: Visualization			**Pages 1 – 65**
1.1	**STATISTICS (Title Slide)**	Course introduction under Visualization	**Page 1**
1.2	**ONE WAY TABLE**	Definition of data; Data Table (rows and columns); Individuals vs. Variables; Quantitative vs. Qualitative variables; Counter-question test; Table orientation	**Pages 2 – 10**
1.3	**BAR GRAPH**	Constructing a frequency table from raw categorical data; Plotting counts on vertical axes with bars; Comparing visual analysis vs. tabular inspection	**Pages 11 – 20**
1.4	**PIE CHART**	Using frequency counts to compute percentage occurrences; Slicing a 100% circle proportionally to represent fruit distribution	**Pages 21 – 26**
1.5	**LINE GRAPH**	Deriving line graphs from bar tops; Tracking continuous time-series trends (monthly expenses); When to choose a line graph over a bar graph	**Pages 27 – 34**
1.6	**O-GIVES**	Constructing cumulative frequency tables from monthly expense data	**Page 35**
1.7	**TWO WAY TABLE**	Independent vs. dependent variables; Breaking multi-year totals (2020–2023) by month; Two-counter-question test; Grouped bar graphs; Dot plot introduction	**Pages 36 – 47**
1.8	**RELATIVE FREQUENCY TABLE**	Converting two-way count tables into percentages; Column-related relative frequencies vs. row-related relative frequencies	**Pages 48 – 53**
1.9	**JOINT DISTRIBUTION**	Comparing two variable distributions simultaneously using row and column percentages (Weight Lost vs. Total Hours Workout)	**Pages 54 – 56**

#	Heading / Topic	Key Subtopics & Concepts Covered	Page Range
1.10	**HISTOGRAM / HISTOGRAMS**	Grouping continuous data into buckets/bins; Plotting bin frequencies; Calculating data range and determining bin widths on real datasets	**Pages 57 – 65**
Module 2: Analysing Data			**Pages 66 – 112**
2.1	**MEASURE OF CENTRAL TENDANCY & OUTLINE**	Section transition from visual charts to mathematical analysis; Describing the center of a dataset	**Pages 66 – 68**
2.2	**Mean**	Average calculation; Statistical formula ($\bar{x} = (1/N) \sum x_i$); Mean as a balancing point on a seesaw	**Pages 69 – 72**
2.3	**Median**	Finding the middle value for odd vs. even numbers of data points	**Pages 73 – 74**
2.4	**Mode**	Identifying most frequent values; Datasets with no mode; Bimodal distributions	**Pages 75 – 77**
2.5	**MEASURE OF SPREAD**	Introduction to data dispersion around the center	**Pages 78 – 80**
2.6	**Range**	Calculating maximum minus minimum values with practical examples	**Pages 81 – 83**
2.7	**Interquartile Range (IQR)**	Percentiles and quartiles (Q1, Q2, Q3, Q4); Formula IQR = Q3 − Q1; Interpreting low vs. high IQR	**Pages 84 – 91**
2.8	**OUTLIERS**	Impact of extreme values on mean vs. median and ML regression models; Outlier removal using Upper Fence (Q3 + 1.5 × IQR) and Lower Fence (Q1 − 1.5 × IQR)	**Pages 92 – 105**
2.9	**5 NUMBER SUMMARY**	Minimum, Q1, Median (Q2), Q3, Maximum; Removing outliers prior to calculation; Constructing and reading a Box Plot	**Pages 106 – 112**
Module 3: Data Distribution			**Pages 113 – 168**
3.1	**INTRODUCTION**	Spread around the mean; Progression from Histogram to Frequency Polygon to Density Curve; Overview of Normal Distribution	**Pages 113 – 116**
3.2	**MEAN, VARIANCE AND STANDARD DEVIATION**	Population vs. Sample concepts; Step-by-step Variance (σ^2) and Standard Deviation (σ) formulas and worked calculations	**Pages 117 – 127**

#	Heading / Topic	Key Subtopics & Concepts Covered	Page Range
3.3	**DENSITY CURVE**	Deriving frequency polygons and smooth density curves; Area under the curve (100%); Symmetrical (Normal) vs. Left- and Right-Skewed distributions; Mean/Median/Mode positions and metric selection	**Pages 128 – 142**
3.4	**Z - SCORE**	Normal distribution properties; Empirical Rule (68–95–99.7 rule); Z-Score formula and interpretation; Using the Z-Table to find cumulative percentages	**Pages 143 – 157**
3.5	**COVARIANCE AND CORRELATION**	Covariance direction (Positive vs. Negative) and formula Cov(X,Y); Pearson Correlation coefficient (r) formula and scale (−1 to +1)	**Pages 158 – 168**
Module 4: Probability and Permutation & Combination			**Pages 169 – 270**
4.1	**PROBABILITY**	Likelihood of events; Coin, card, and dice examples; Simple probability formula P(H); Experimental vs. theoretical probability	**Pages 169 – 177**
4.2	**UNION VS INTERSECTION & ADDITION RULE**	Venn diagrams; General Addition Rule P(A ∪ B) = P(A) + P(B) − P(A ∩ B); Mutually exclusive events; Two-way contingency table probability	**Pages 178 – 187**
—	***Unpopulated / Blank Pages***	*Remaining slides in the file contain no text or visual content*	***Pages 188 – 270***